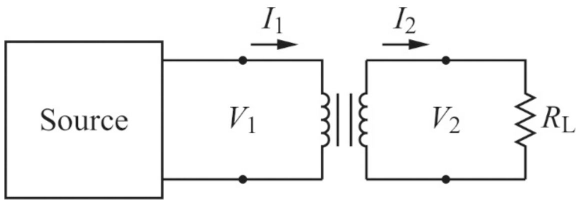
Exercise (4) – Problem from Chapter 2

14. An audio signal generator has an output impedance of 600 Ω. To drive an 8 Ω speaker with maximum power transfer, an impedance-matching transformer is used between the generator and the speaker. What is the necessary turns-ratio $n_{21} = \frac{N_2}{N_1}$ for such a transformer?



Solution:

For maximum power transfer, the primary side ***effective resistance*** must be equal to the output impedance of the generator, $R_{\text{eff}} = R_{\text{gen}}$. The output current and voltage are given by $V_2 = I_2 R_L$. Expressing the secondary side voltage and current with the primary side voltage and current, respectively, we get:

$$V_2 = \frac{N_2}{N_1} V_1 = I_2 R_{\text{L}} = \frac{N_1}{N_2} I_1 R_{\text{L}} \quad \rightarrow \quad V_1 = I_1 \left(\frac{N_1^2}{N_2^2} R_{\text{L}} \right) = I_1 R_{\text{eff}} \quad \rightarrow \quad R_{\text{eff}} = \frac{N_1^2}{N_2^2} R_{\text{L}} = \frac{R_{\text{L}}}{n_{21}^2}$$

With the given parameters, the turns-ratio:

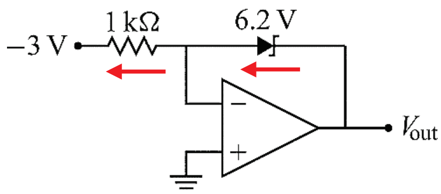
$R_{eff} := 600\ \Omega$ $\qquad$ $R_L := 8\ \Omega$ $\qquad$ $\boxed{n_{21} := \sqrt{\frac{R_L}{R_{eff}}} = 0.115}$

Exercise (4) – Problem from Chapter 6

8. Determine the output voltage for circuit **(c)** shown in the figures.

Solution:

(c) By the first op-amp law, the voltage at the (−) input is 0 V. Thus, the voltage across the resistor is 3 V producing a current of 3 mA through the 1 kΩ going to the left. By the second law, all of this goes through the Zener. The reverse direction of the current requires that the Zener is in breakdown mode, so the voltage across it is 6.2 V. Since the voltage at the (−) input is 0 V:



(c)

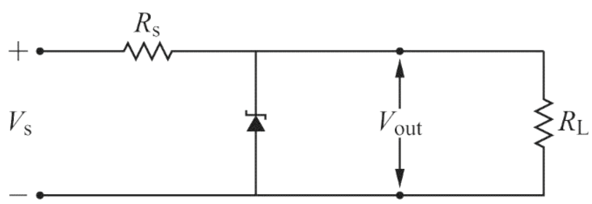
$\boxed{V_o = 6.2\ \text{V}}$

Exercise (4) – Problems from Chapter 3

7. Let V_s = 30 V, R_s = 300 Ω, and the Zener breakdown voltage be V_b = 15 V in the circuit shown. Suppose we vary the load resistor R_L in order to vary the load current through R_L. Plot the output voltage of the regulator as a function of load current from 0 to 75 mA. Over what load current range is the regulator effective?

Solution:

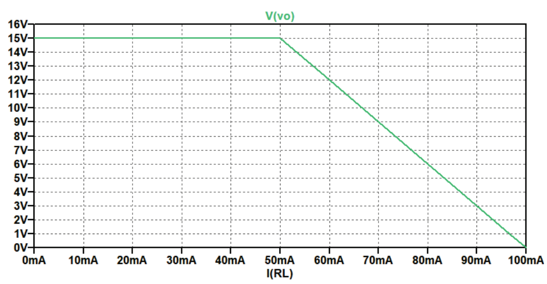
As in *Problem-3.6(a)*, for proper Zener operation, we must satisfy that $V_{\text{out}} > V_{\text{b}}$, and, hence $R_{\text{L}} > R_{\text{L,min}}$. Then, we should find the maximum load current below which the regulator is effective, $I_{\text{L}} < I_{\text{L,max}}$:



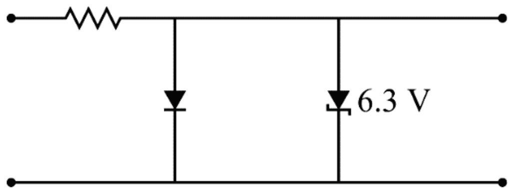
$V_s := 30\ \text{V}$ $\qquad$ $R_s := 300\ \Omega$ $\qquad$ $V_b := 15\ \text{V}$

$$R_{L\_min} := R_s \cdot \frac{V_b}{V_s - V_b} = 300\ \Omega$$

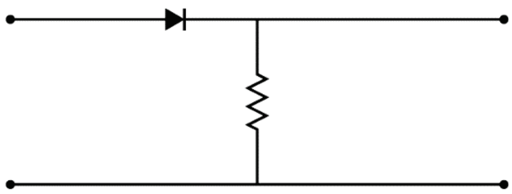
$$\boxed{I_L < I_{L\_max} := \frac{V_b}{R_{L\_min}} = 50\ \text{mA}}$$



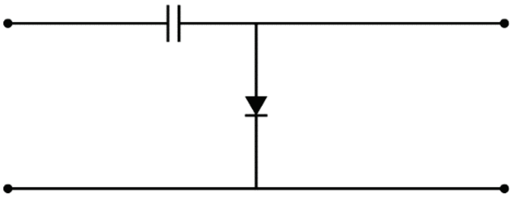
2. Sketch the output waveforms expected when a 100 Hz, 5 V_p sine wave is applied to each of the circuits below. Specify important voltage levels and time scales. The input is on the left and the output is on the right.



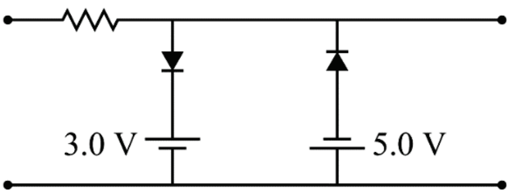
Asymmetric clipper (+0.6 V … −6.3 V).



Half-wave rectifier.

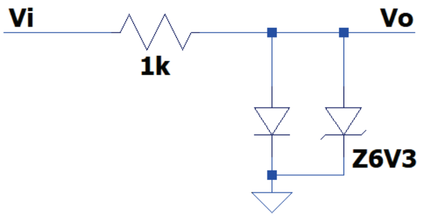


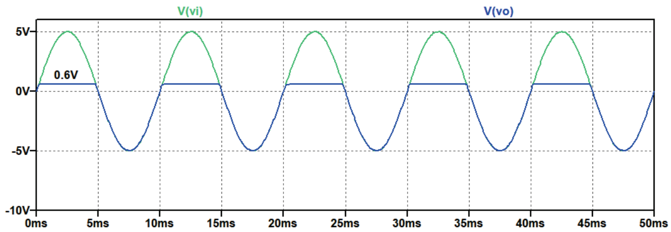
Diode clamp (+0.6 V … $-V_p$+0.6 V).

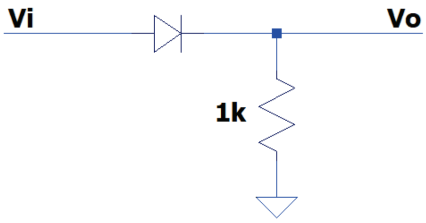


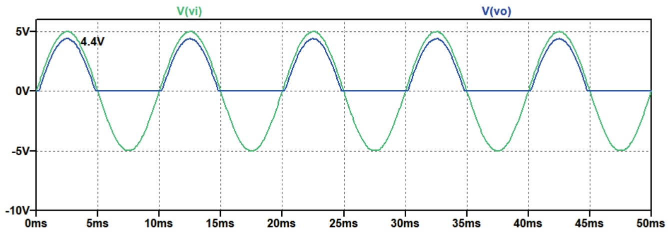
Asymmetric clipper (+3 V … −5 V).

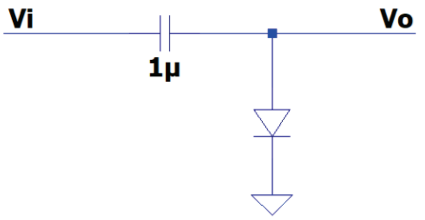
Solution:

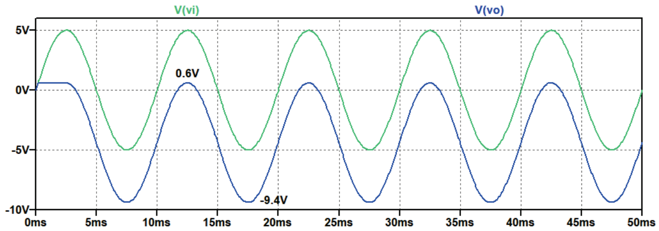


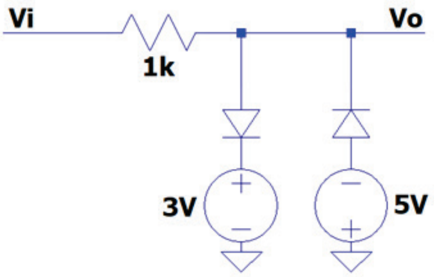


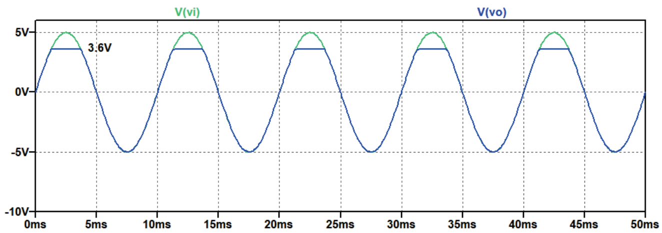












8. Design a variable voltage power supply capable of supplying 250 mA with output voltage range 5 to 15 V. Specify the relevant details (e.g., values, model numbers, power rating) for all the components you use.

Number	Type	V_{out}(V)	I_{out}(A)	V_{drop}(V)	P_{max}(W)
LM7805CT	Fixed	+5	1.5	30	1.7
LM7815CT	Fixed	+15	1.5	20	1.7
LM7905CT	Fixed	−5	1.5	25	1.7
LM7915CT	Fixed	−15	1.5	30	1.7
LM317T	Adjustable	1.2 to 37	1.5	40	20
LM337T	Adjustable	−1.2 to −37	1.5	40	15

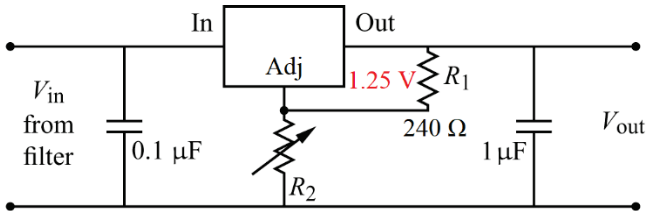
Name	V_{DC}	Ripple factor r
Simple RC	$V_p - \frac{I_{DC}}{2fC}$	$\frac{1}{2\sqrt{3}fR_L C}$
RC π-section	$V_p - \left(\frac{1}{2fC_1} + R\right) I_{DC}$	$\frac{1}{4\pi\sqrt{3}f^2 C_1 C_2 R R_L}$

Solution:

Since we need a variable output voltage, we are directed to the variable voltage regulator circuit shown. From the Table, we choose the ***LM317 positive adjustable regulator*** which can output the necessary voltage range and specified current. The regulator requires an input voltage at least 3 V above its highest output voltage so:

$V_{out\_max} := 15 \text{ V} \qquad \Delta V_{io\_min} := 3 \text{ V}$

$V_{in\_min} := V_{out\_max} + \Delta V_{io\_min} = 18 \text{ V}$



I_{OUT}	PARAMETER	LM317	UNIT
1.5 A	Input voltage range	4.25 - 40	V
	Load regulation accuracy	1.5	%
	PSRR (120 Hz)	64	dB
	Recommended operating temperature	0 to 125	°C

		MIN	MAX	UNIT
V_O	Output voltage	1.25	37	V
$V_I - V_O$	Input-to-output differential voltage	3	40	V
I_O	Output current	0.01	1.5	A

PARAMETER	MIN	TYP	MAX	UNIT
ADJUST terminal current		50	100	µA
Reference voltage	1.2	1.25	1.3	V
Minimum load current to maintain regulation		3.5	10	mA

We choose the resistor values associated with the voltage regulator. From the datasheet, the controlling equation is the following (with V_{REF} being the ***reference voltage*** between pins OUT and ADJ and neglecting the small adjust current, I_{adj}):

$$V_{out} = V_{REF}\left(1 + \frac{R_2}{R_1}\right) + I_{adj}R_2 \approx V_{REF}\left(1 + \frac{R_2}{R_1}\right)$$

Choosing the recommended R_1 to maintain minimum load current and solving for R_2:

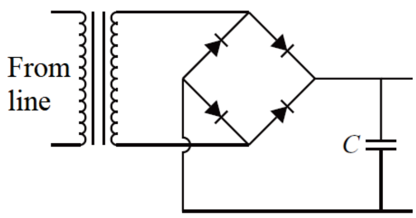
$$R_2 = R_1 \frac{V_{out} - V_{REF}}{V_{REF}}$$

$V_{REF} := 1.25 \text{ V} \qquad R_1 := 240\ \Omega \qquad I_{out\_min} := \frac{V_{REF}}{R_1} = 5.208 \text{ mA} \qquad V_{out\_min} := 5 \text{ V}$

$R_{2\_min} := R_1 \cdot \frac{V_{out\_min} - V_{REF}}{V_{REF}} = 720\ \Omega \qquad R_{2\_max} := R_1 \cdot \frac{V_{out\_max} - V_{REF}}{V_{REF}} = 2640\ \Omega$

Using a potentiometer with standard value of 2.2 kΩ in series with a fixed 510 Ω resistor will cover this range.

Next, we may choose a simple capacitor filter, so we have: $V_{in,min} = V_p - \frac{V_{AC,pp}}{2} = V_p - \frac{I_L}{2Cf_r}$. This gives a DC level in the middle of the ripple, so we'll double the second term to get the minimum of the signal. Also V_p is the peak of the rectified signal so we add the diode voltage to get the peak of the transformer output. Then, using the maximum (worst-case) load current and set the input filter capacitor, the **50 Hz** mains transformer peak amplitude for a ***full-wave bridge rectifier*** must be:



$V_d := 0.6 \text{ V} \qquad I_L := 250 \text{ mA} \qquad f_r := 100 \text{ Hz} \qquad C := 630\ \mu\text{F}$

$V_{p\_tr} := V_{in\_min} + 2 \cdot V_d + \frac{2 \cdot I_L}{2 \cdot f_r \cdot C} = 23.168 \text{ V}$

Note, that a standard 24 V transformer can be used with the selected standard capacitor value. Using these parameters, the ripple factor can be calculated for the input of the regulator:

$R_{L\_min} := \frac{V_{in\_min}}{I_L} = 72\ \Omega \qquad r_{max} := \frac{1}{2 \cdot \sqrt{3} \cdot f_r \cdot R_{L\_min} \cdot C} = 6.364\ \%$

This ripple is significant, so we may select a larger capacitor. However, we can also make use of the regulator extreme large power supply rejection ratio of $PSSR = 64$ dB:

$$pssr := 10^{\frac{PSSR}{-20\,\text{dB}}} = 0.063\,\% \qquad r_{out} := r_{max} \cdot pssr = 0.004\,\%$$

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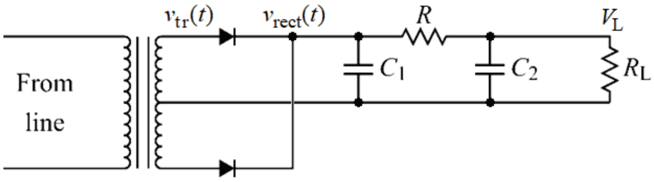
4. Determine the necessary values of the components (R, C_1, and C_2, and V_p for the **60 Hz** mains transformer output) for the circuit, with the requirements that the load voltage be 15 V with 0.01% or less ripple and the load current be 100 mA.

Solution:



From the specifications we can work backwards and determine the component values. We calculate R_L, then from r, the C_1C_2R product at the 120 Hz ripple frequency:

$$f_r := 120\,\text{Hz} \qquad V_L := 15\,\text{V} \qquad I_L := 100\,\text{mA} \qquad r := 10^{-4}$$

$$R_L := \frac{V_L}{I_L} = 150\,\Omega \qquad C1C2R := \frac{1}{r \cdot 4 \cdot \pi \cdot \sqrt{3} \cdot f_r^{\,2} \cdot R_L} = 212.704\,\text{mF mF}\,\Omega$$

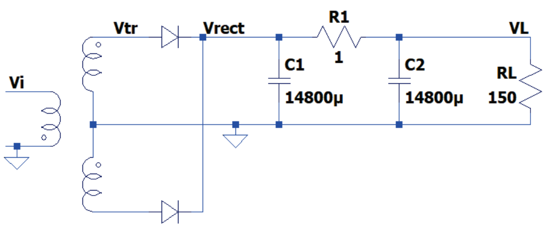
Also, from the equation for V_L, we see that for good regulation we want large C_1 and C_2, and small R, so choose the values and check the requirements:

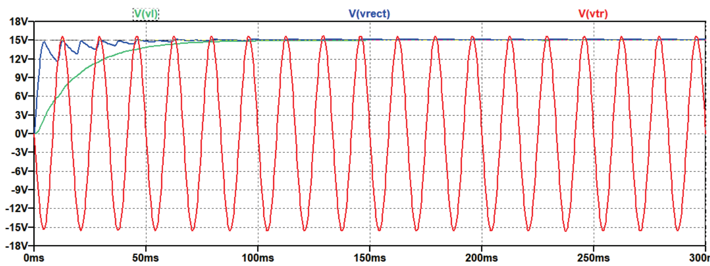
$$R := 1\,\Omega \qquad C_1 := 14800\,\mu\text{F} \qquad C_2 := C_1$$

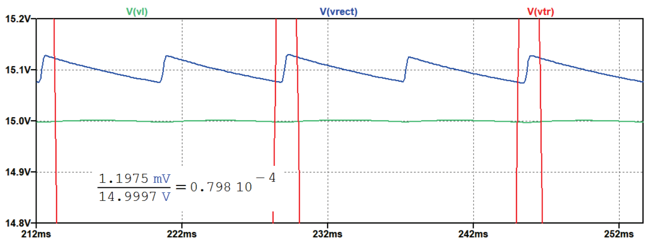
$$C_1 \cdot C_2 \cdot R = 219.04\,\text{mF mF}\,\Omega \qquad R + R_L = 151\,\Omega \qquad 2 \cdot \pi \cdot f_r \cdot C_2 \cdot R \cdot R_L = 1673.841\,\Omega$$

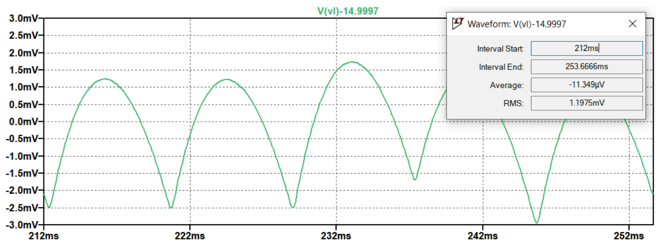
Using these values in the expression for V_L and solving for V_p gives (without and with the diode voltage drop):

$$\boxed{V_p := V_L + I_L \cdot \left(R + \frac{1}{2 \cdot C_1 \cdot f_r}\right) = 15.128\,\text{V} \qquad V_p + V_d = 15.728\,\text{V}}$$









<u>Side note (only for demonstration)</u>: derivation of DC voltage and ripple factor:

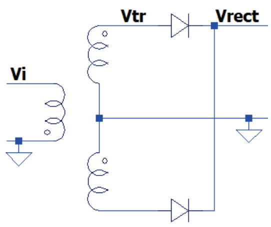
Definition of ripple factor: $r = \frac{V_{\mathrm{AC},rms}}{V_{\mathrm{DC}}}$

The DC component is defined as the ***time-average*** or ***mean*** of the signal within one period:

$$V_{\mathrm{DC}} = \frac{1}{T}\int_0^T v_{\mathrm{rect}}(t)dt$$

The effective voltage (displayed on an AC voltmeter) is given as the root-mean-square (***rms***) of the voltage signal:

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^T v_{\mathrm{rect}}^2(t)dt}$$



The AC component is the ***zero-mean time-varying*** component of the signal:

$$v_{\mathrm{AC}}(t) = v_{\mathrm{rect}}(t) - V_{\mathrm{DC}} \quad \rightarrow \quad V_{\mathrm{AC},rms} = \sqrt{V_{rms}^2 - V_{\mathrm{DC}}^2}$$

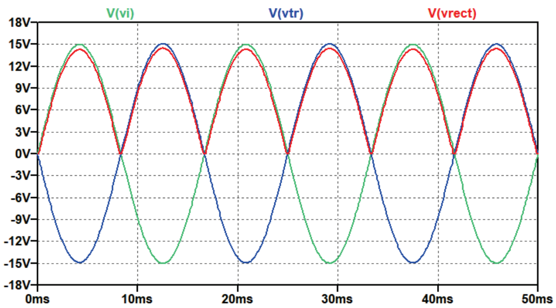
The transformer output is ***center-tapped***, the diodes form a ***full-wave rectifier***. Without any filtering, the DC and *rms* values can be calculated for period $T/2$ as:

$$V_{\mathrm{DC}} = \frac{2}{T}\int_0^{T/2} V_p \sin\left(\frac{2\pi t}{T}\right)dt = \frac{2}{T}\int_0^{\pi} V_p \sin(\theta)\frac{d\theta}{(2\pi/T)} = \frac{V_p}{\pi}[-\cos\theta]_0^{\pi} = \frac{2V_p}{\pi}$$

$$V_{rms}^2 = \frac{2}{T}\int_0^{T/2} V_p^2 \sin^2\left(\frac{2\pi t}{T}\right)dt = \frac{V_p^2}{\pi}\int_0^{\pi}\sin^2\theta\, d\theta = \frac{V_p^2}{\pi}\int_0^{\pi}\frac{1}{2}[1-\cos(2\theta)]d\theta = \frac{V_p^2}{2} - 0$$

$$V_{\mathrm{AC},rms} = V_p\sqrt{\frac{1}{2} - \frac{4}{\pi^2}}$$

$$r := \frac{V_{AC\_rms}}{V_{DC}} = 48.343\ \%$$



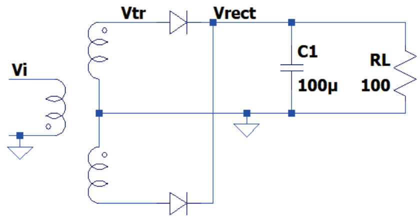
Considering the diode forward voltage (e.g. $V_d = 0.6$ V):

$$v_{\mathrm{rect}}(t) = v_{\mathrm{tr}}(t) - V_d$$

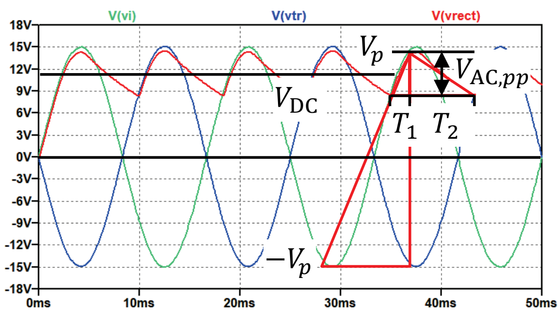
$$V_{\mathrm{DC,rect}} = V_{\mathrm{DC,tr}} - V_d$$

The circuit employs an RC π-filter that can be decomposed into a capacitor and an RC filter.

The expression for the ripple factor of the capacitor filter output can be approximated by a sawtooth waveform. During time T_1, the diodes conduct charging the capacitor up to the peak rectifier voltage V_p. During time T_2, the rectifier voltage drops below the peak voltage and the capacitor discharges through the load. Since the charge-discharge cycle occurs for each half-cycle for a full-wave rectifier, the period of the rectified waveform is:



$$\frac{T}{2} = T_1 + T_2 \quad \rightarrow \quad T_1 = \frac{T}{2} - T_2$$



Using similar triangles, we can approximate T_1:

$$\frac{V_{\mathrm{AC},pp}}{T_1} \approx \frac{2V_p}{T/2} \quad \rightarrow \quad T_1 \approx \frac{T}{4}\frac{V_{\mathrm{AC},pp}}{V_p}$$

The DC and AC voltages are given by:

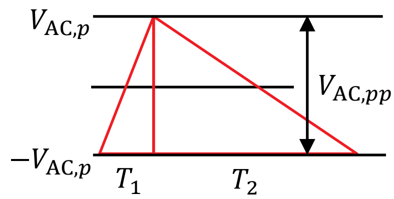
$$V_{\mathrm{DC}} = V_p - \frac{V_{\mathrm{AC},pp}}{2} \quad \rightarrow \quad V_{\mathrm{AC},pp} = 2(V_p - V_{\mathrm{DC}})$$

Now the charge and discharge periods: $T_1 \approx \frac{T}{4}\frac{2(V_p - V_{\mathrm{DC}})}{V_p} = \frac{T}{2} - \frac{T}{2}\frac{V_{\mathrm{DC}}}{V_p} \quad \rightarrow \quad T_2 = \frac{T}{2}\frac{V_{\mathrm{DC}}}{V_p}$

We can express the discharge (or load) current, and the AC component using $f = \frac{1}{T}$ and the ripple frequency $f_r = 2f$:

$$I_{\mathrm{L}} = C_1\frac{\Delta V}{\Delta t} = C_1\frac{V_{\mathrm{AC},pp}}{T_2} \quad \rightarrow \quad V_{\mathrm{AC},pp} = \frac{I_{\mathrm{L}}}{C_1}T_2 = \frac{I_{\mathrm{L}}}{C_1}\frac{T}{2}\frac{V_{\mathrm{DC}}}{V_p} = \frac{I_{\mathrm{L}}}{C_1}\frac{1}{2f}\frac{V_{\mathrm{DC}}}{V_p} = \frac{I_{\mathrm{L}}}{C_1 f_r}\frac{V_{\mathrm{DC}}}{V_p}$$

Now, find the *rms* value from the *pp* value for the sawtooth ripple (being symmetric around the DC voltage):



$$v_{\mathrm{AC}}(t) = \begin{cases} \frac{V_{\mathrm{AC},pp}}{T_1}t, & -\frac{T_1}{2} \le t < \frac{T_1}{2} \\ -\frac{V_{\mathrm{AC},pp}}{T_2}t, & -\frac{T_2}{2} \le t < \frac{T_2}{2} \end{cases}$$

$$V_{\mathrm{AC},rms}^2 = \frac{1}{T}\int_{\frac{-T_1}{2}}^{\frac{T_1}{2}} \frac{V_{\mathrm{AC},pp}^2}{T_1^2}t^2 dt + \frac{1}{T}\int_{\frac{-T_2}{2}}^{\frac{T_2}{2}} \left(-\frac{V_{\mathrm{AC},pp}}{T_2}t\right)^2 dt = \frac{1}{T}\frac{V_{\mathrm{AC},pp}^2}{T_1^2}\frac{2T_1^3}{8\cdot 3} + \frac{1}{T}\frac{V_{\mathrm{AC},pp}^2}{T_2^2}\frac{2T_2^3}{8\cdot 3} = \frac{V_{\mathrm{AC},pp}^2}{T}\frac{T_1+T_2}{4\cdot 3}$$

$$V_{\mathrm{AC},rms} = \frac{V_{\mathrm{AC},pp}}{2\sqrt{3}} = \frac{I_{\mathrm{L}}}{2\sqrt{3}C_1 f_r}\frac{V_{\mathrm{DC}}}{V_p} = \frac{I_{\mathrm{L}}}{2\sqrt{3}C_1 f_r}\frac{V_{\mathrm{DC}}}{V_p}, \qquad I_{\mathrm{L}} = \frac{V_{\mathrm{DC}}}{R_{\mathrm{L}}}, \qquad \frac{V_{\mathrm{DC}}}{V_p} = \frac{V_{\mathrm{DC}}}{V_{\mathrm{DC}} + V_{\mathrm{AC},p}} = \frac{1}{1 + \frac{V_{\mathrm{AC},pp}}{2V_{\mathrm{DC}}}}$$

The ripple factor for the capacitor filter is then:

$$r = \frac{V_{\mathrm{AC},rms}}{V_{\mathrm{DC}}} = \frac{1}{2\sqrt{3}C_1 f_r R_{\mathrm{L}}}\frac{V_{\mathrm{DC}}}{V_p} = \frac{1}{2\sqrt{3}C_1 f_r R_{\mathrm{L}}}\frac{1}{1 + \frac{\sqrt{3}\,V_{\mathrm{AC},rms}}{V_{\mathrm{DC}}}} = \frac{1}{2\sqrt{3}C_1 f_r R_{\mathrm{L}}}\left(\frac{1}{1+\sqrt{3}\,r}\right)$$

As r decreases, the term in bracket approaches unity.

It is possible to further reduce the amount of ripple across the filter capacitor by using an additional RC filter section with the purpose to pass most of the DC component while attenuating as much of the AC component as possible. As the ripple component of the capacitor filter is much smaller than the DC component, the operation of the RC circuit can be analysed using ***superposition*** for the DC and AC components of signal.

The DC equivalent circuit, where both capacitors are open-circuit for DC operation, are the series connection of the filter and load resistors. Thus DC output of the RC filter stage is given by the resistor-divider:

$$V_{\mathrm{L}} = V_{\mathrm{DC}}\frac{R_{\mathrm{L}}}{R + R_{\mathrm{L}}} = V_{\mathrm{DC}} - I_{\mathrm{L}}R = V_p - \frac{V_{\mathrm{AC},pp}}{2} - I_{\mathrm{L}}R = V_p - I_{\mathrm{L}}\left(R + \frac{1}{2C_1 f_r}\right)$$

The AC output of the RC filter stage is given by the divider circuit R and $C_2||R_{\mathrm{L}}$:

$$\tilde{Z}_2 = \frac{R_{\mathrm{L}}\tilde{Z}_C}{R_{\mathrm{L}} + \tilde{Z}_C} = \frac{R_{\mathrm{L}}\frac{1}{j\omega_r C_2}}{R_{\mathrm{L}} + \frac{1}{j\omega_r C_2}} = \frac{R_{\mathrm{L}}}{j\omega_r C_2 R_{\mathrm{L}} + 1}$$

$$\tilde{V}_{\mathrm{ACL}} = \tilde{V}_{\mathrm{AC}}\frac{\tilde{Z}_2}{R + \tilde{Z}_2} = \frac{\frac{R_{\mathrm{L}}}{j\omega_r C_2 R_{\mathrm{L}} + 1}}{R + \frac{R_{\mathrm{L}}}{j\omega_r C_2 R_{\mathrm{L}} + 1}} = \frac{R_{\mathrm{L}}}{R(j\omega_r C_2 R_{\mathrm{L}} + 1) + R_{\mathrm{L}}} = \frac{R_{\mathrm{L}}}{(j\omega_r C_2 R_{\mathrm{L}} R) + (R + R_{\mathrm{L}})}$$

$$V_{\mathrm{ACL},rms} = V_{\mathrm{AC},rms}\frac{R_{\mathrm{L}}}{\sqrt{(R + R_{\mathrm{L}})^2 + (\omega_r C_2 R_{\mathrm{L}} R)^2}} \approx V_{\mathrm{AC},rms}\frac{1}{\omega_r C_2 R}$$

The latter simplification can be made if $R + R_{\mathrm{L}} \ll \omega_r C_2 R_{\mathrm{L}} R$, yielding the ripple factor for the RC π-filter:

$$r = \frac{V_{\mathrm{ACL},rms}}{V_{\mathrm{L}}} = \frac{1}{2\sqrt{3}C_1 f_r R_{\mathrm{L}}}\frac{1}{\omega_r C_2 R} = \frac{1}{4\pi\sqrt{3}f_r^2 C_1 C_2 R R_{\mathrm{L}}}$$